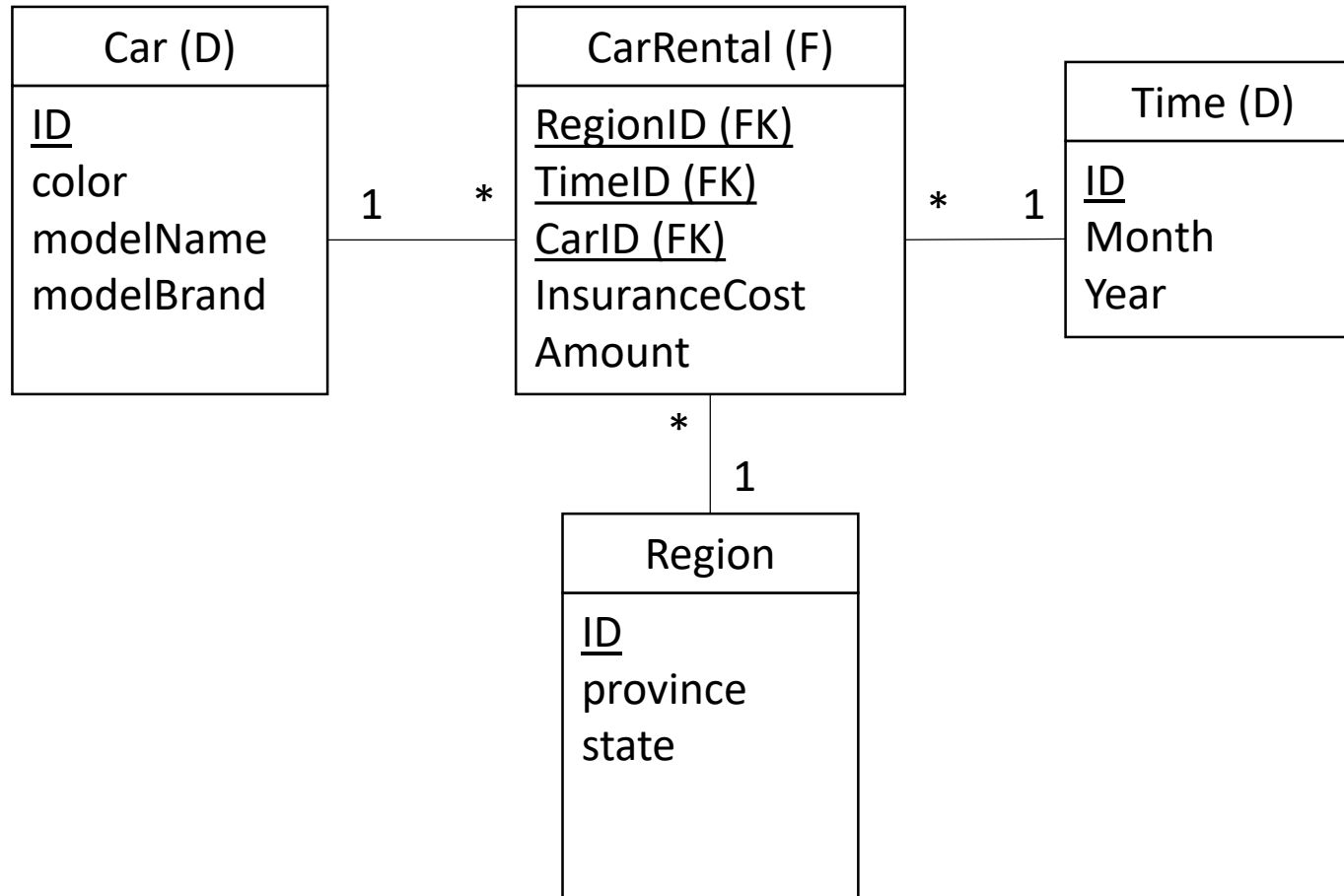


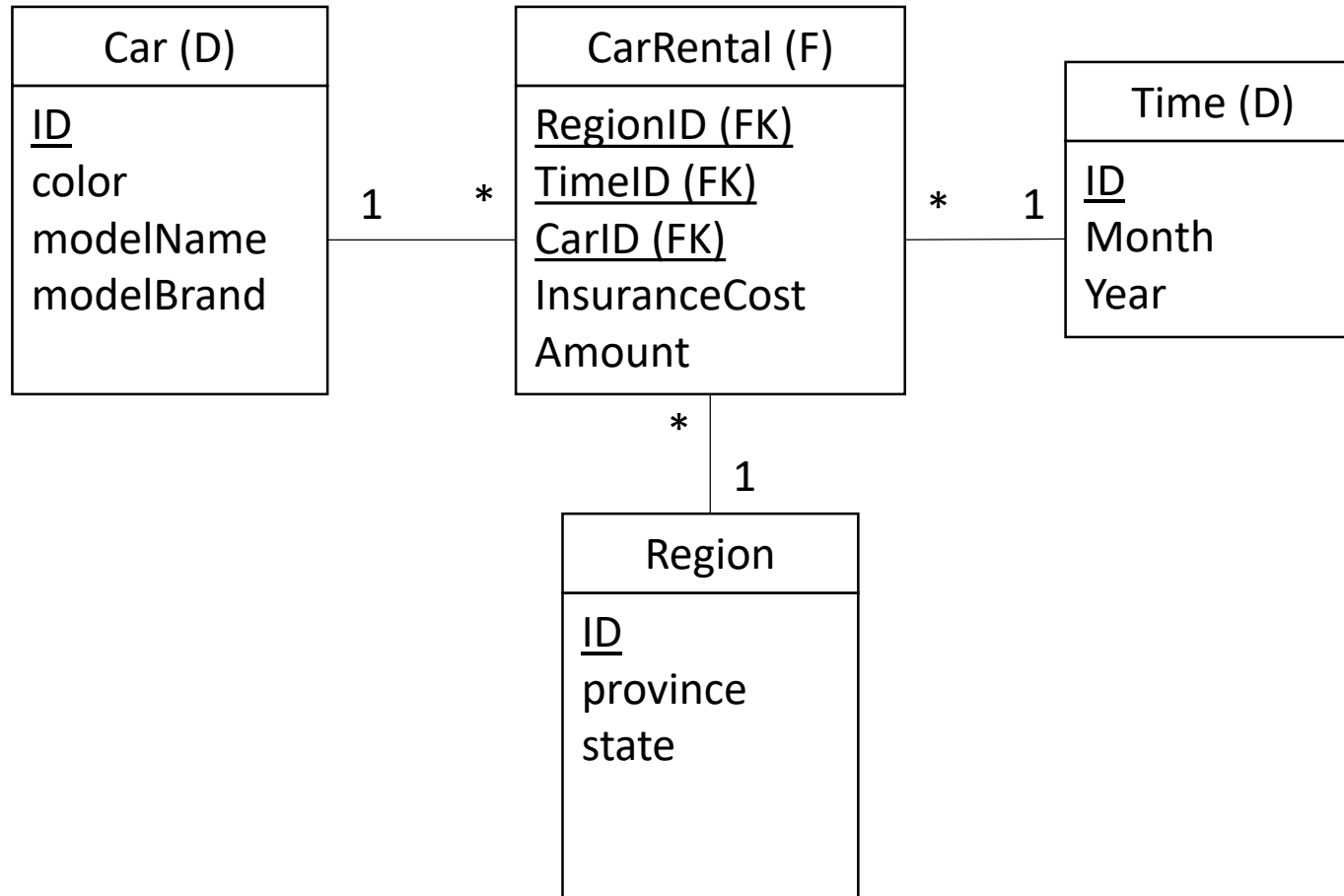
Multidimensional Translation Example

Data cube UML design and transformations



A := Roll-up(CarRental, Region.state, SUM)
B := Drill-down(A, CarRental, Region.province)
C := Roll-up(B, Car.AllCars, SUM)
D := ChangeBase(C, {Time, Region})
E := Roll-up(D, Time.AllDays, SUM)
F := ChangeBase(E, {Region})
G := Selection(F, province = 'Girona')
H := Projection(G, amount)

Data cube UML design and transformations

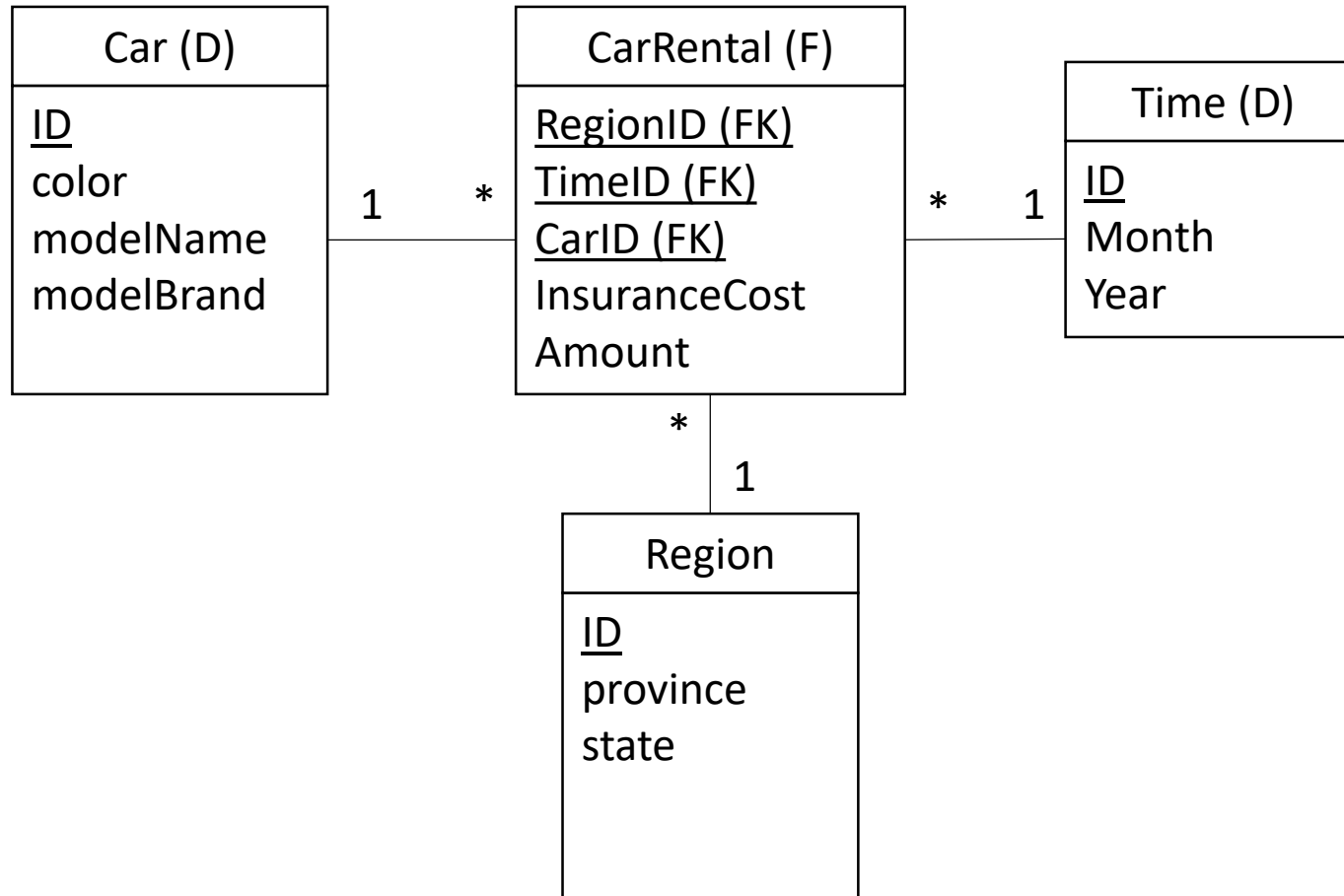


We start by building the CarRental cube

We apply the pattern:

```
SELECT dimensionsIDs, factAttributes
FROM FactTable, DimensionTables
WHERE join dimensionIDs with FactTable
GROUP BY dimensionsIDs
ORDER BY dimensionsIDs
```

Data cube UML design and transformations

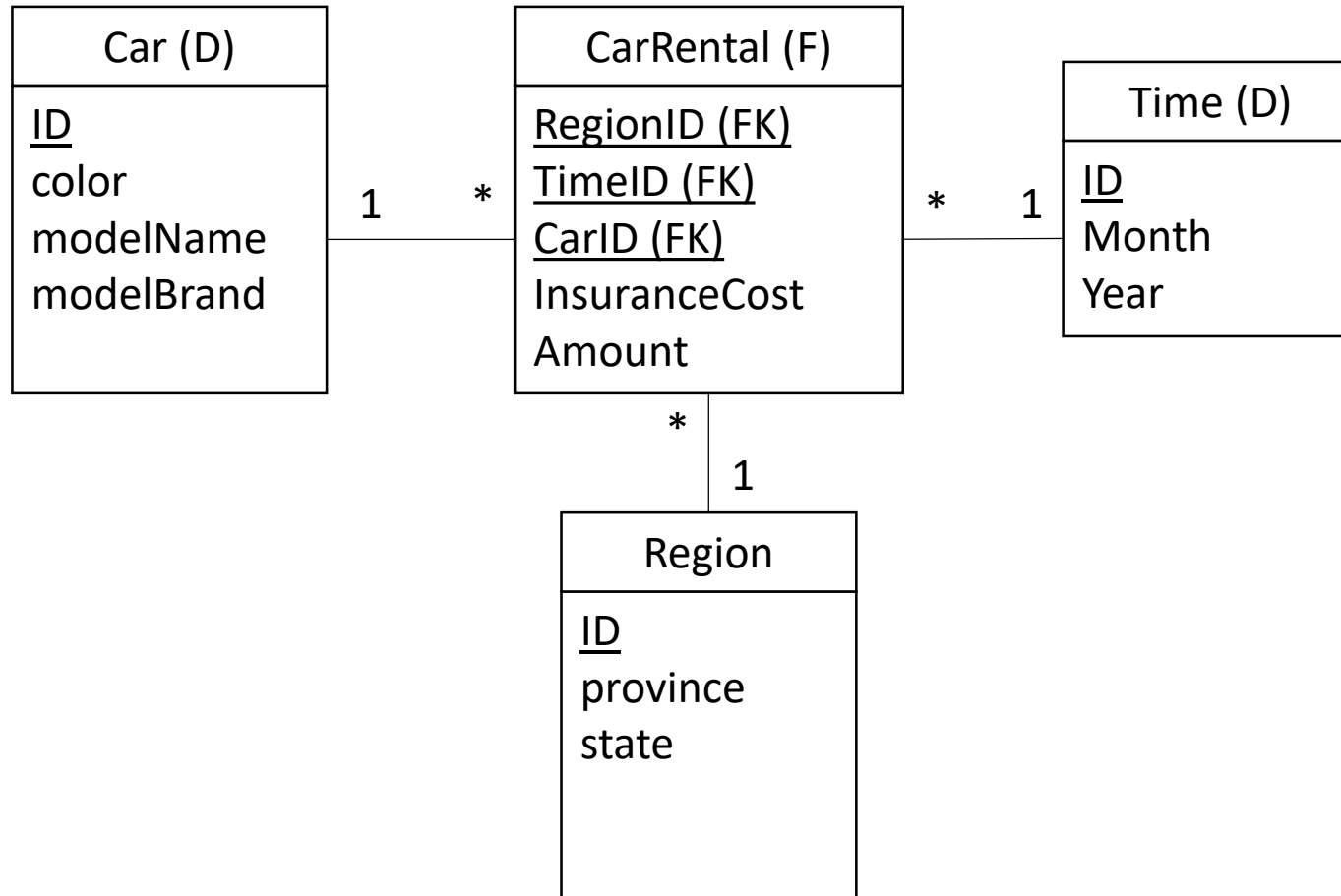


We start by building the CarRental cube

We apply the pattern:

```
SELECT dimensionsIDs, factAttributes
FROM CarRental CR,
      Car C,
      Time T,
      Region R
WHERE join dimensionIDs with FactTable
GROUP BY dimensionsIDs
ORDER BY dimensionsIDs
```

Data cube UML design and transformations

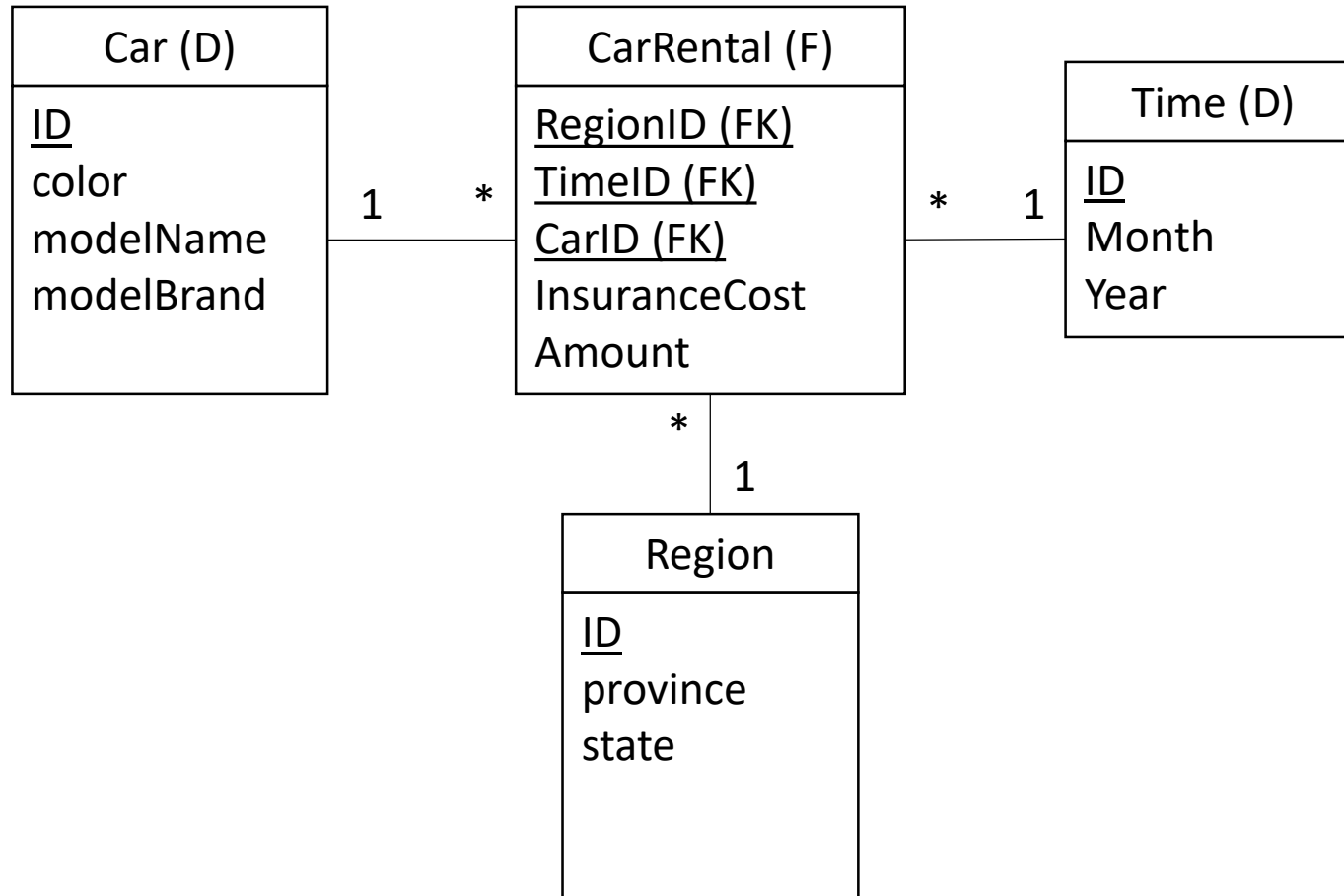


We start by building the CarRental cube

We apply the pattern:

```
SELECT dimensionsIDs, factAttributes
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY dimensionsIDs
ORDER BY dimensionsIDs
```

Data cube UML design and transformations

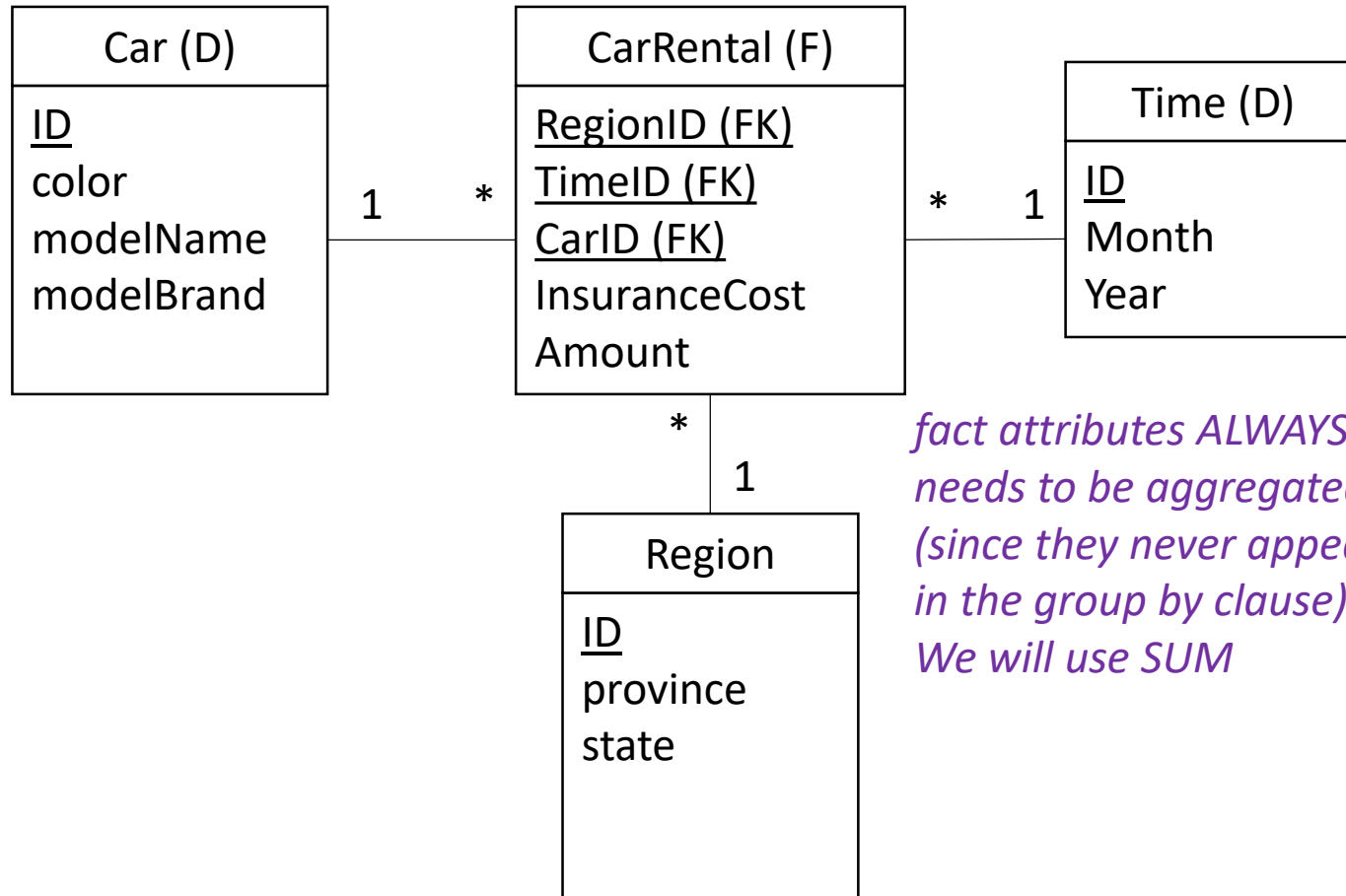


We start by building the CarRental cube

We apply the pattern:

```
SELECT dimensionsIDs, factAttributes
FROM CarRental CR,
      Car C,
      Time T,
      Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY C.ID, T.ID, R.ID
ORDER BY C.ID, T.ID, R.ID
```

Data cube UML design and transformations

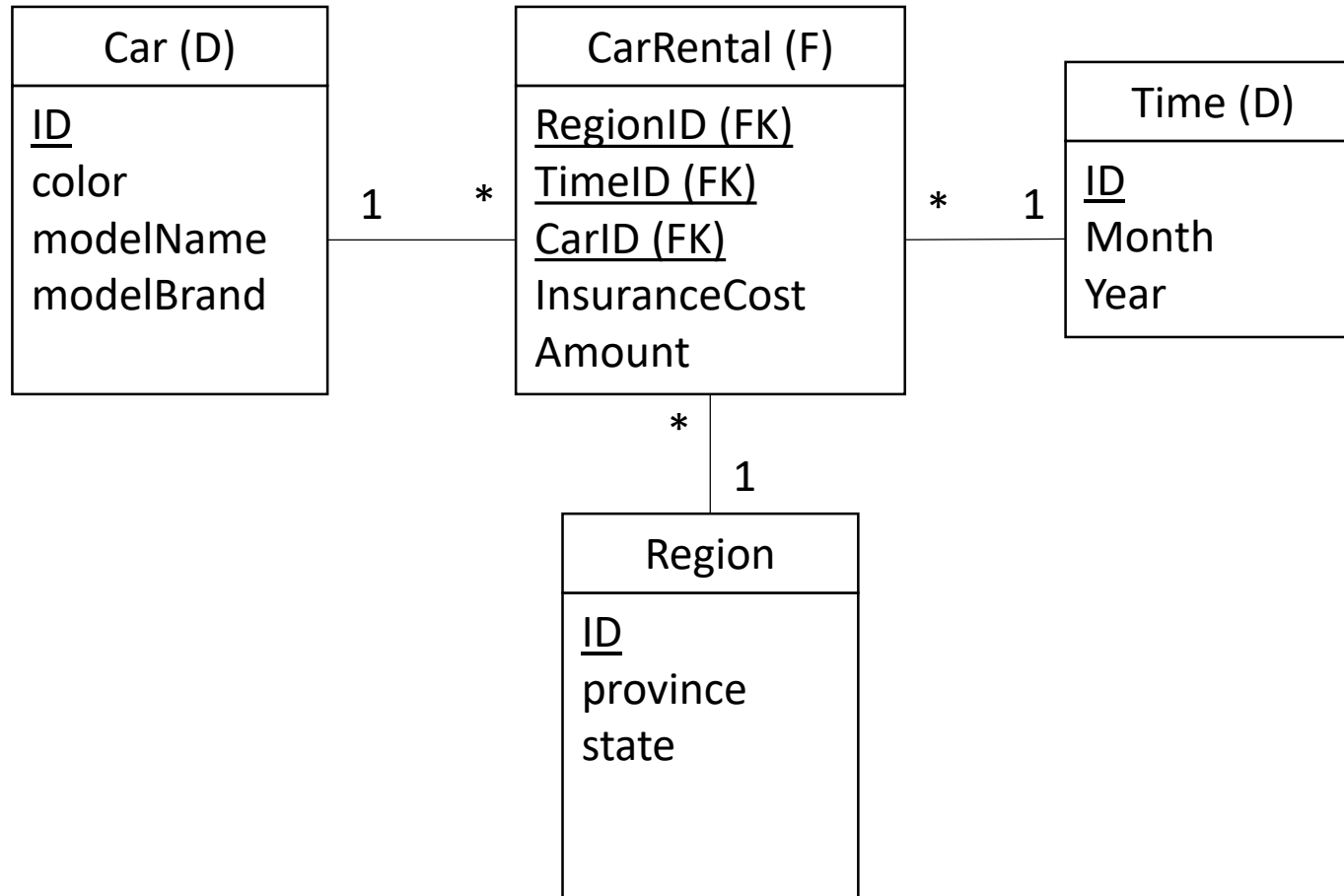


*fact attributes ALWAYS
needs to be aggregated
(since they never appear
in the group by clause)
We will use SUM*

We start by building the CarRental cube

```
SELECT C.ID, T.ID, R.ID,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY C.ID, T.ID, R.ID
ORDER BY C.ID, T.ID, R.ID
```

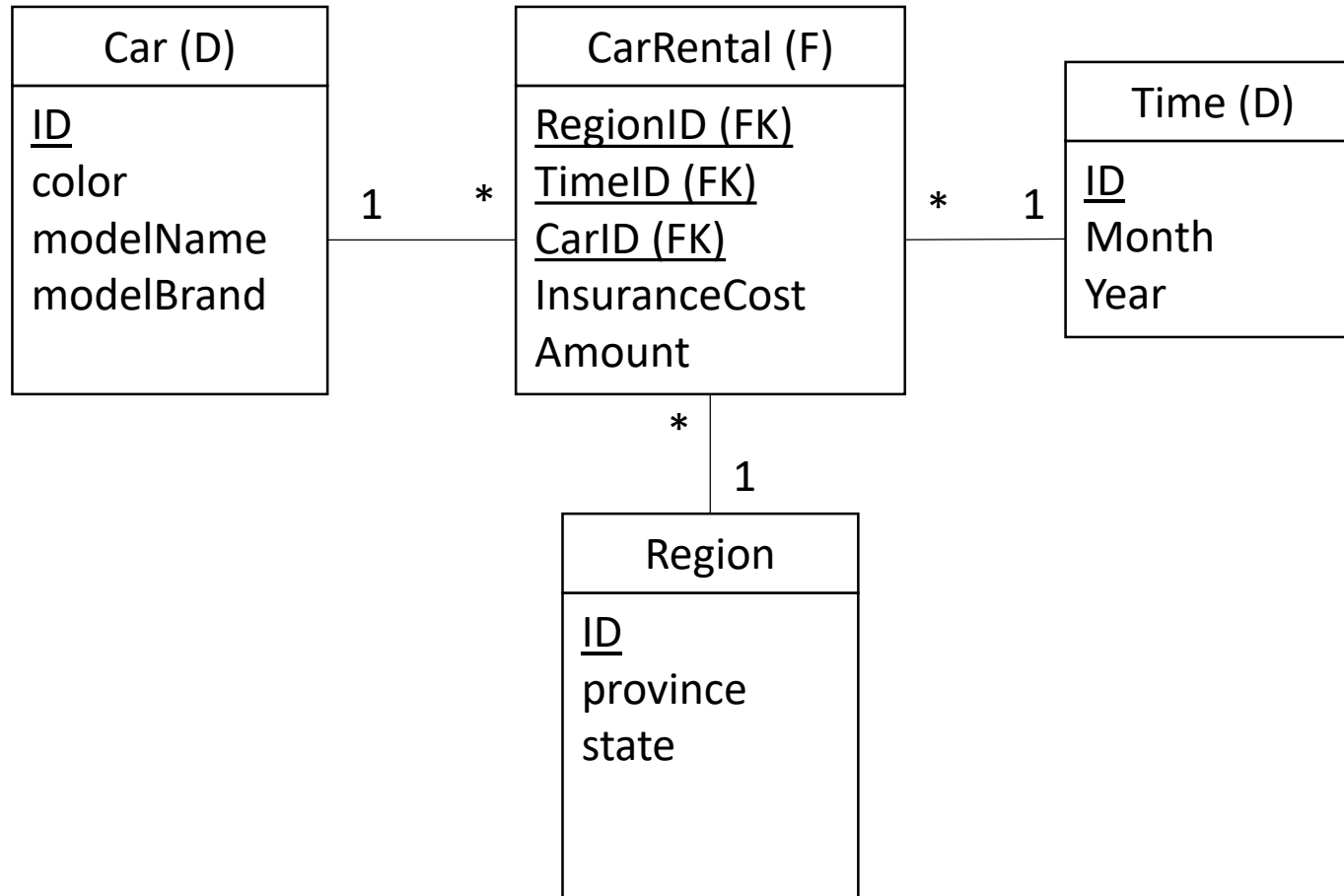
Data cube UML design and transformations



We start by building the CarRental cube
We now modify this query to obtain the different data cubes A,B,C,...

```
SELECT C.ID, T.ID, R.ID,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY C.ID, T.ID, R.ID
ORDER BY C.ID, T.ID, R.ID
```

Data cube UML design and transformations

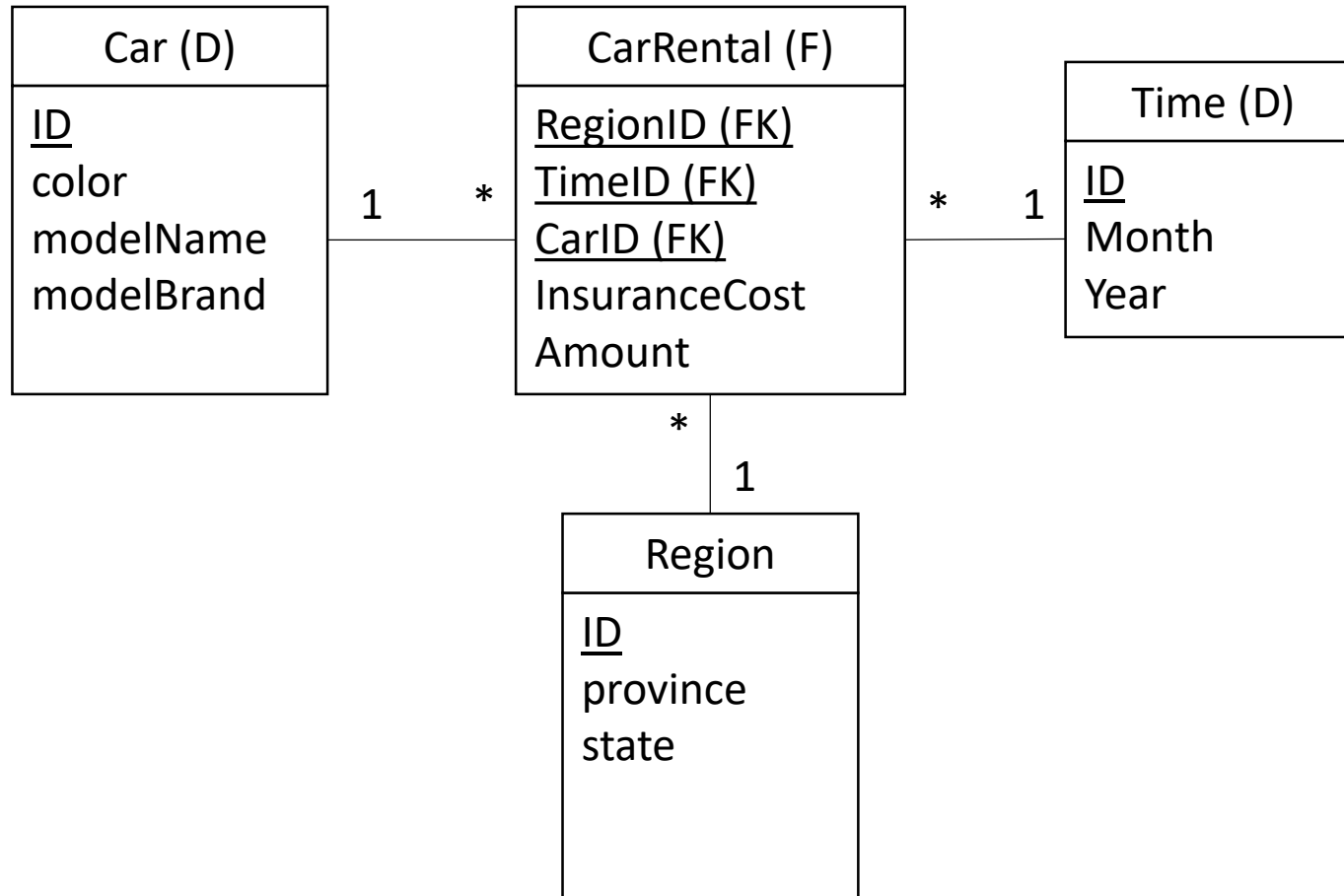


A := Roll-up(CarRental, Region.**state**, SUM)

To apply a roll-up to a given level, just modify the group/order by and select clause to that level

```
SELECT C.ID, T.ID, R.state,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY C.ID, T.ID, R.state
ORDER BY C.ID, T.ID, R.state
```

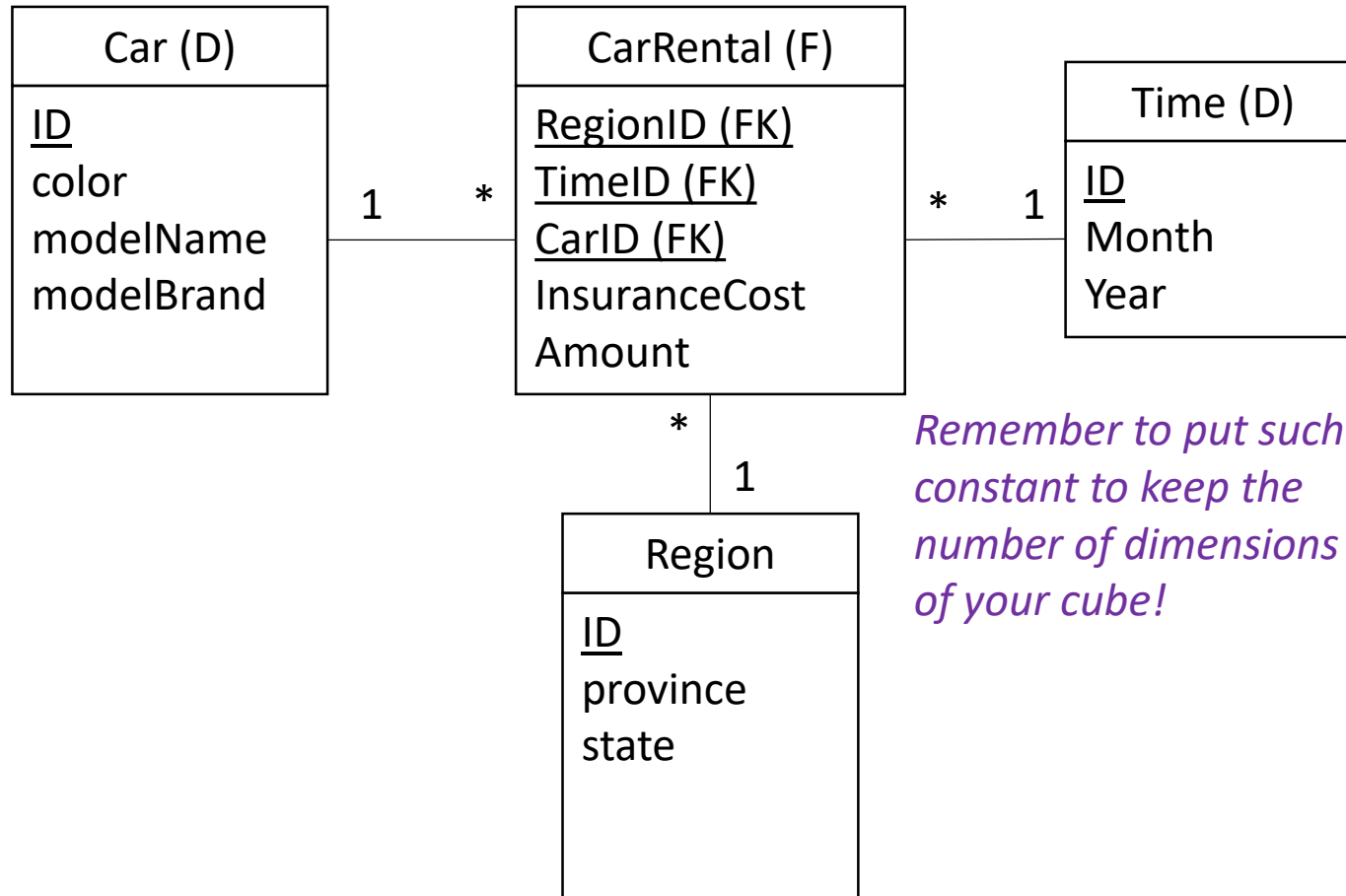
Data cube UML design and transformations



B := Drill-down(A, CarRental, Region.**province**)
To apply a drill-down to a given level, just modify
The group/order by and select clause to that level

```
SELECT C.ID, T.ID, R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY C.ID, T.ID, R.province
ORDER BY C.ID, T.ID, R.province
```

Data cube UML design and transformations



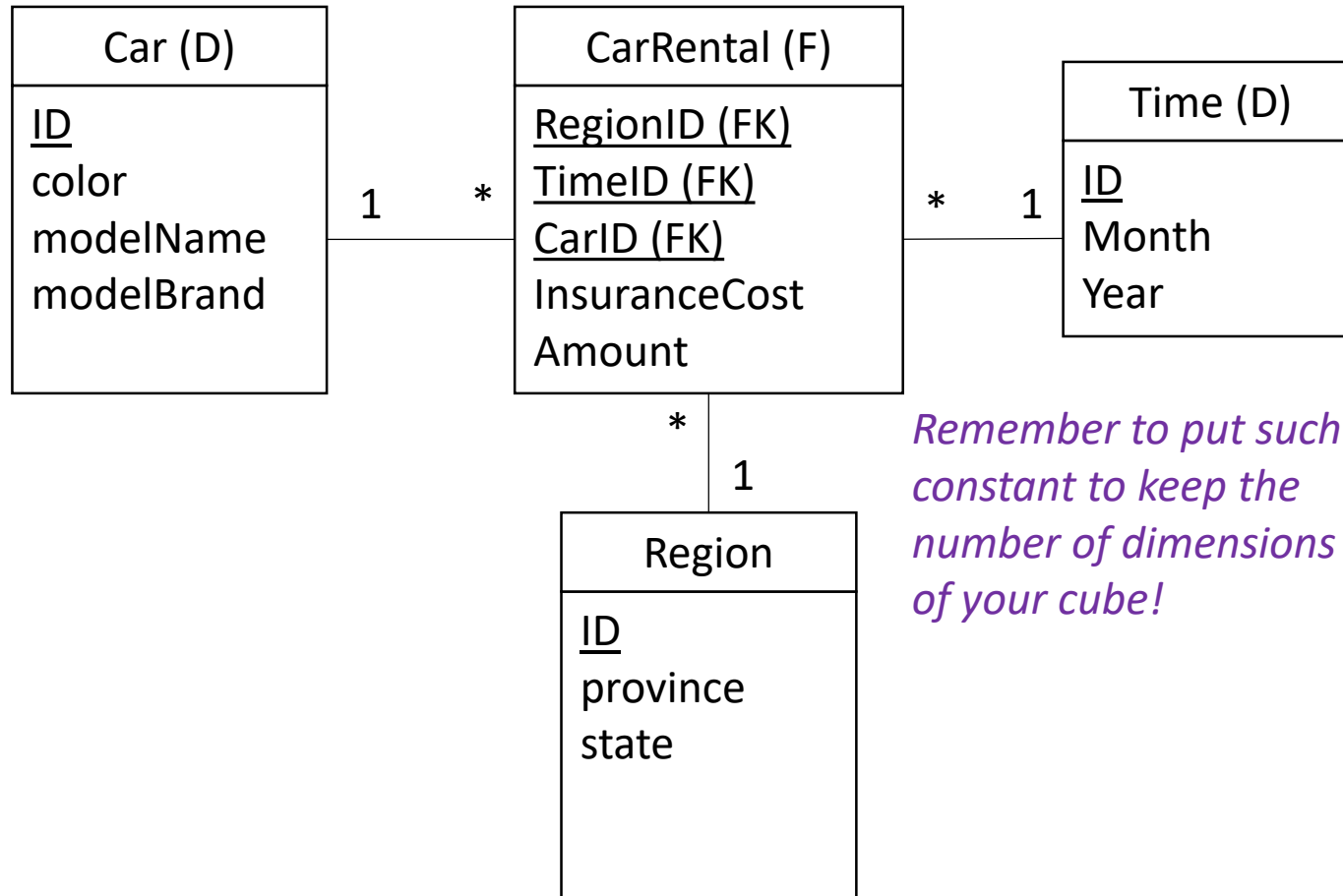
Remember to put such constant to keep the number of dimensions of your cube!

C := Roll-up(B, Car.AllCars, SUM)
 To apply a roll-up to the top level, remove the dimension from the group/order by, and replace the selected dimension for a constant

```

SELECT C.ID 'AllCars', T.ID, R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY C.ID, T.ID, R.province
ORDER BY C.ID, T.ID, R.province
  
```

Data cube UML design and transformations



Remember to put such constant to keep the number of dimensions of your cube!

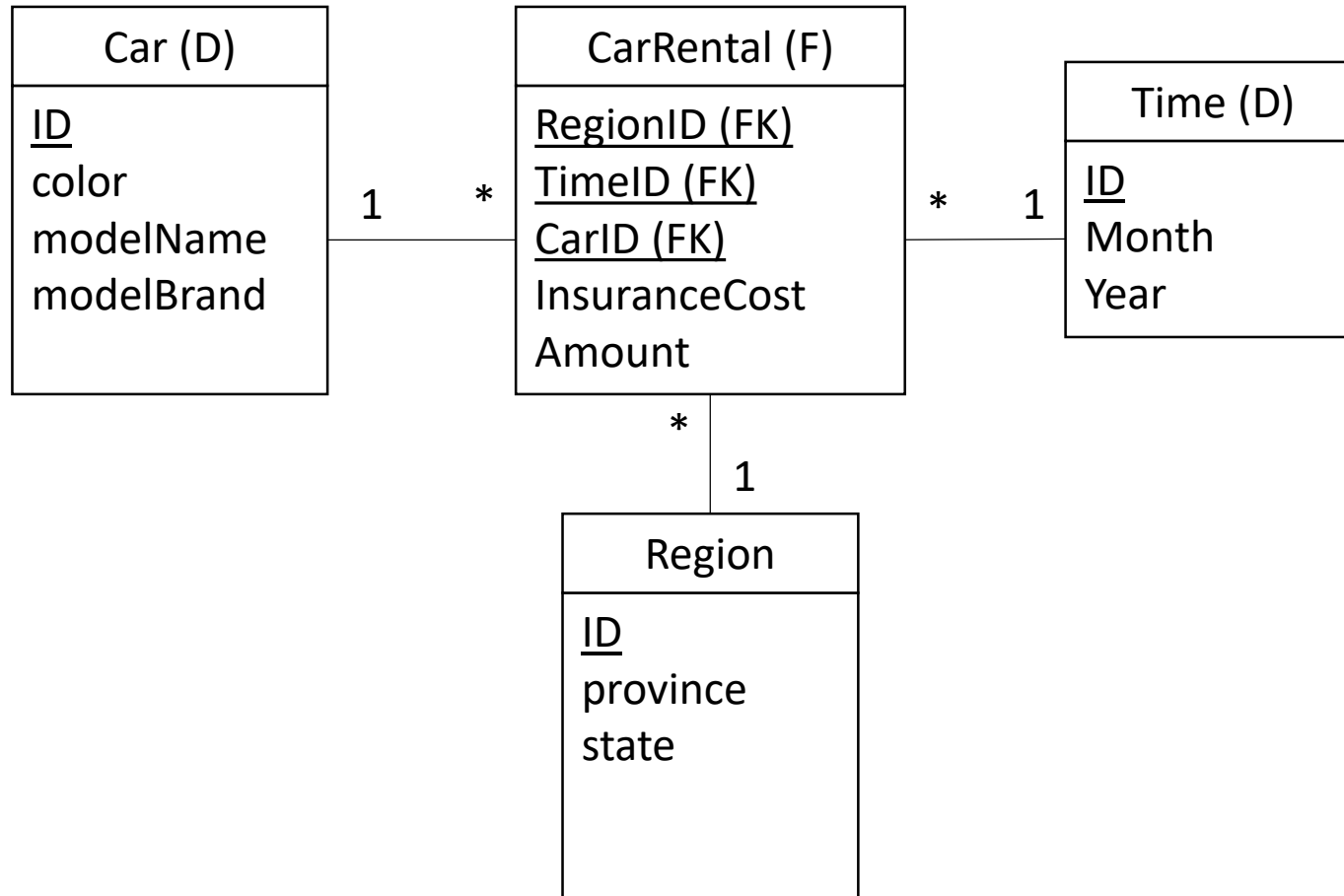
C := Roll-up(B, Car.AllCars, SUM)
 To apply a roll-up to the top level, remove the dimension from the group/order by, and replace the selected dimension for a constant
 SELECT 'AllCars', T.ID, R.province,

SUM(insuranceCost),
 SUM(amount)

FROM CarRental CR,
 Car C,
 Time T,
 Region R

WHERE CR.regionID = R.id AND
 CR.timeID = T.id AND
 CR.carID = Car.id
 GROUP BY T.ID, R.province
 ORDER BY T.ID, R.province

Data cube UML design and transformations

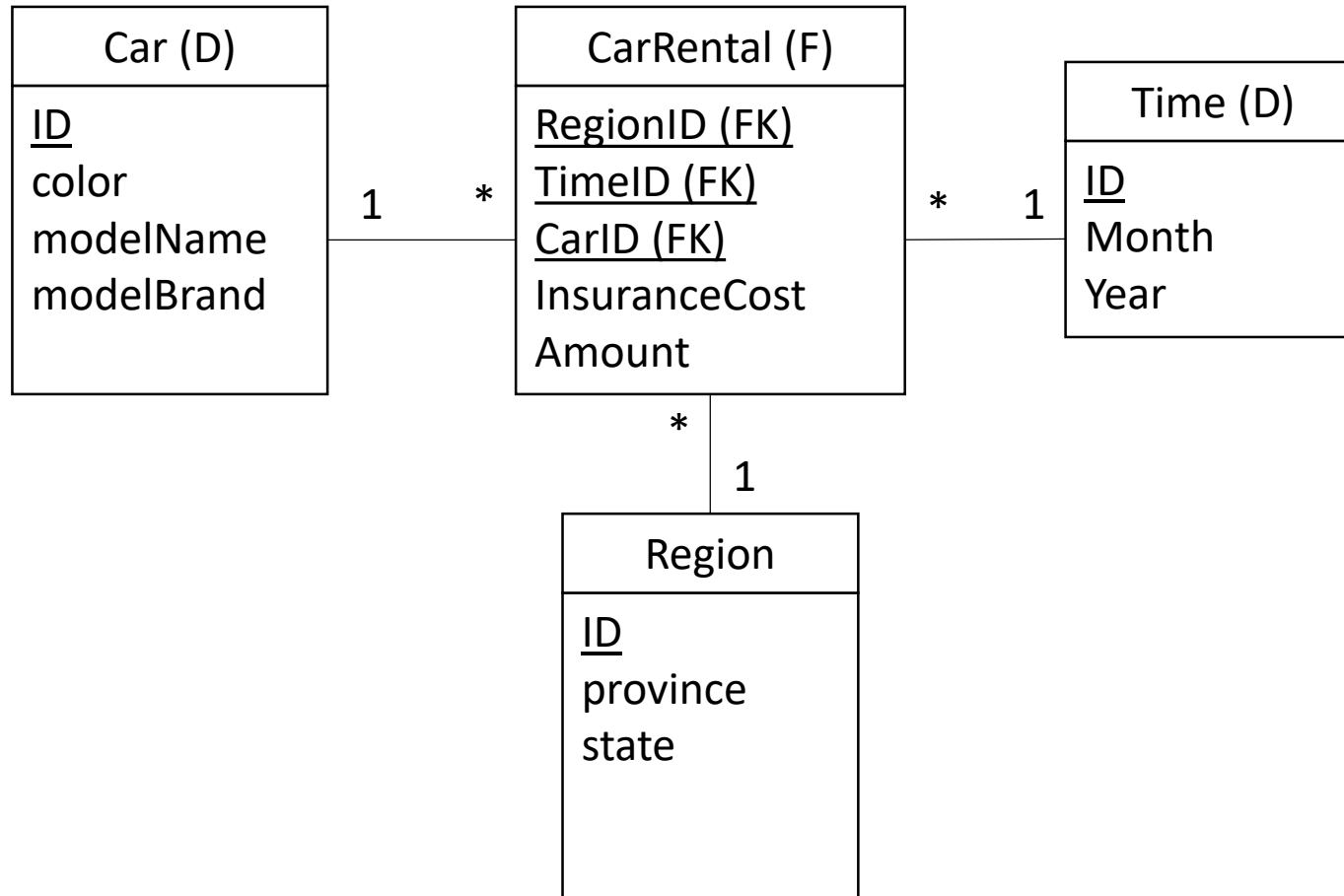


$D := \text{ChangeBase}(C, \{\text{Time}, \text{Region}\})$

The ChangeBase operation might be used to remove dimensions

```
SELECT T.ID, R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY T.ID, R.province
ORDER BY T.ID, R.province
```

Data cube UML design and transformations

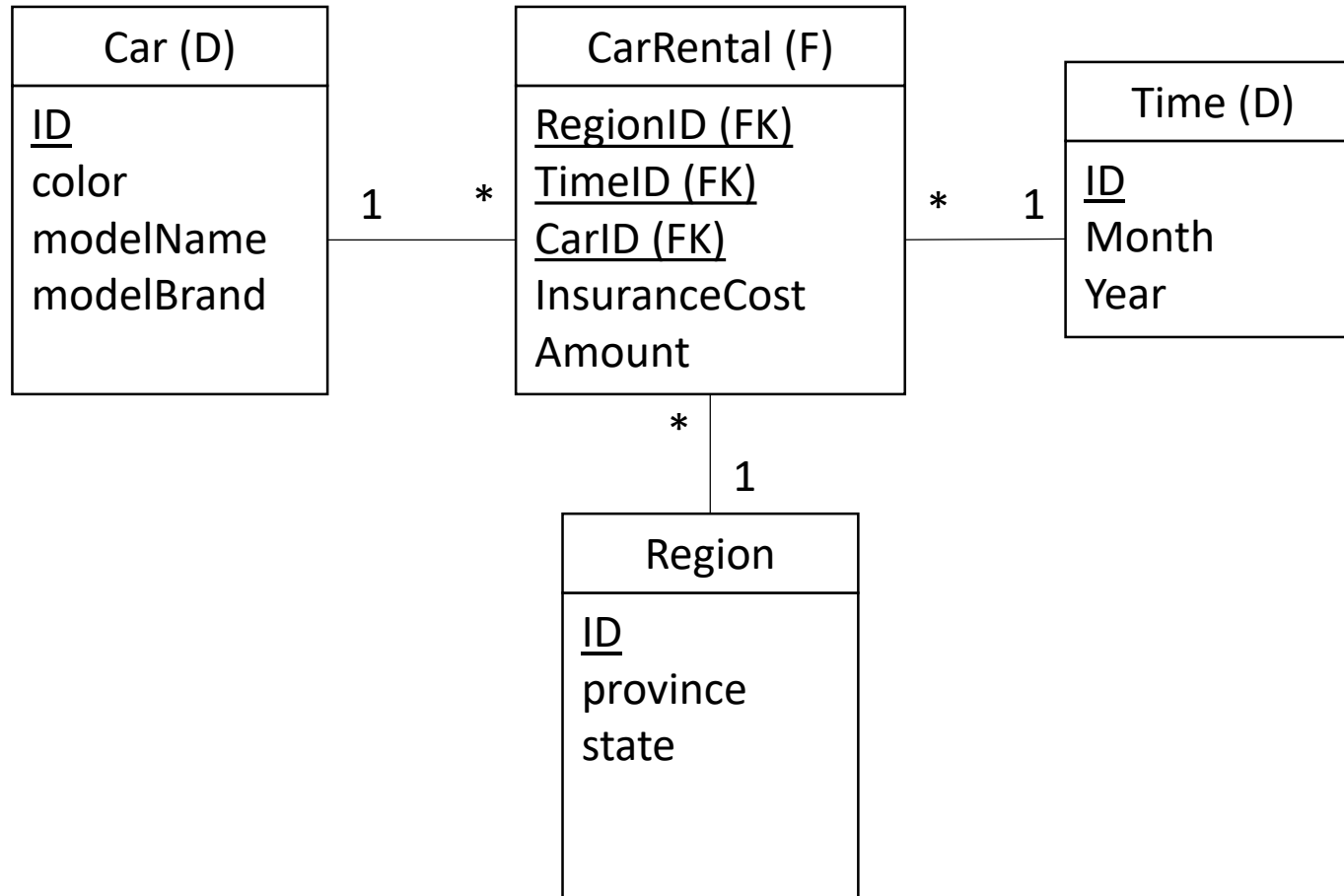


$E := \text{Roll-up}(D, \text{Time.AllDays}, \text{SUM})$

```

SELECT T.ID 'AllDays', R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id
GROUP BY T.ID, R.province
ORDER BY T.ID, R.province
  
```

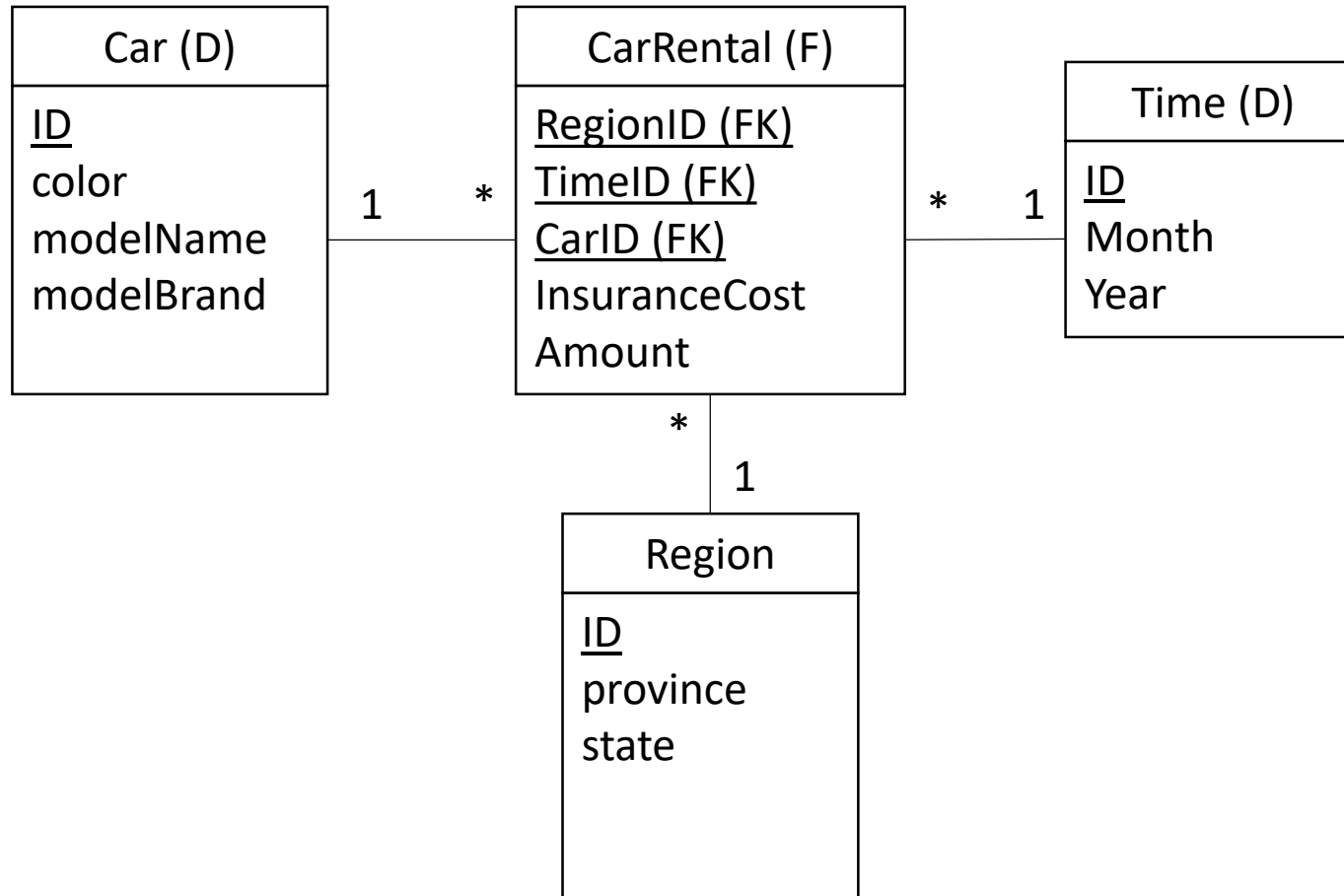
Data cube UML design and transformations



$F := \text{ChangeBase}(E, \{\text{Region}\})$

```
SELECT R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = C.id
GROUP BY R.province
ORDER BY R.province
```

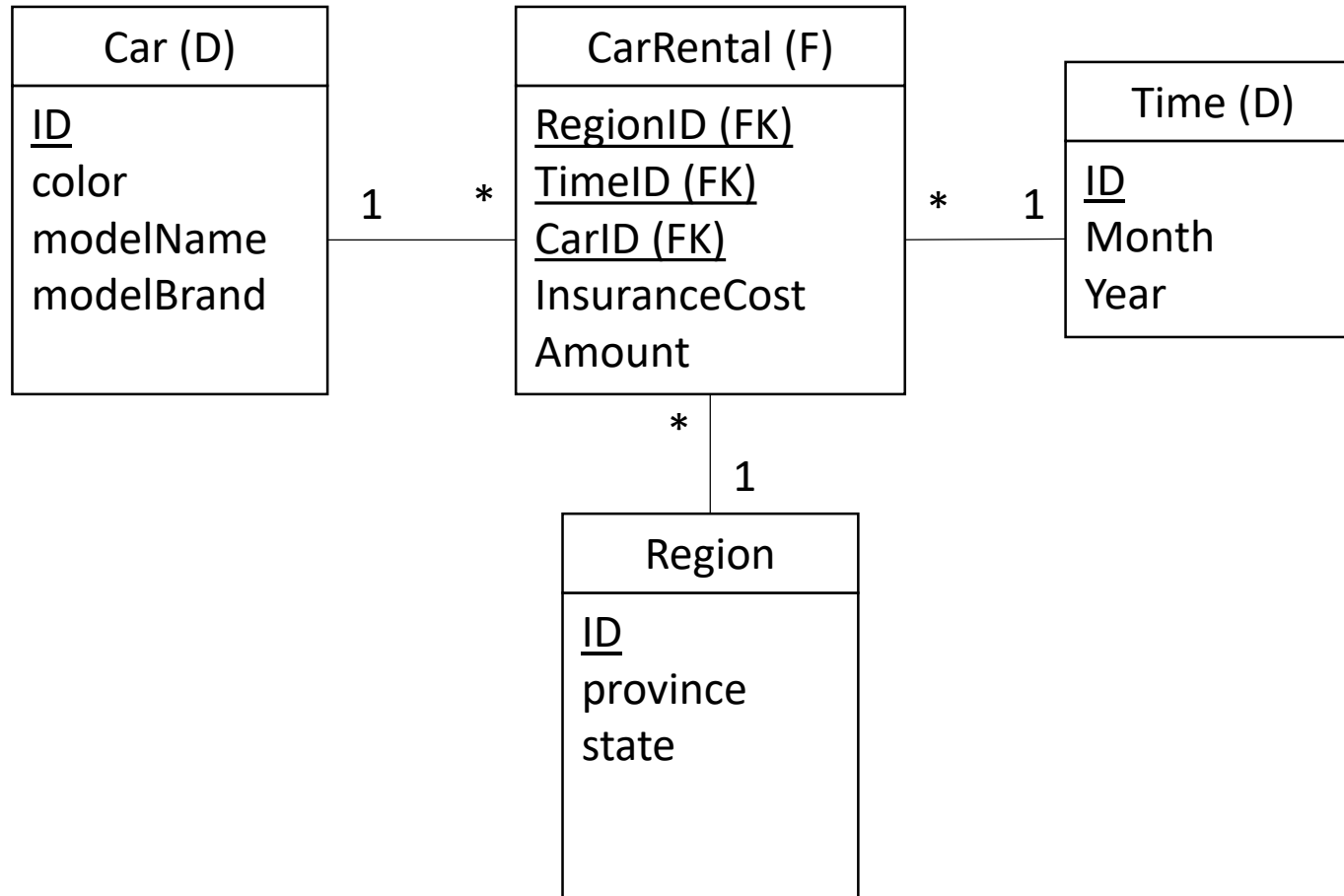
Data cube UML design and transformations



G := Selection(F, province = 'Girona')
Select operations are translated in the where clause

```
SELECT R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id AND
      R.province = 'Girona'
GROUP BY R.province
ORDER BY R.province
```

Data cube UML design and transformations



$H := \text{Projection}(G, \text{amount})$

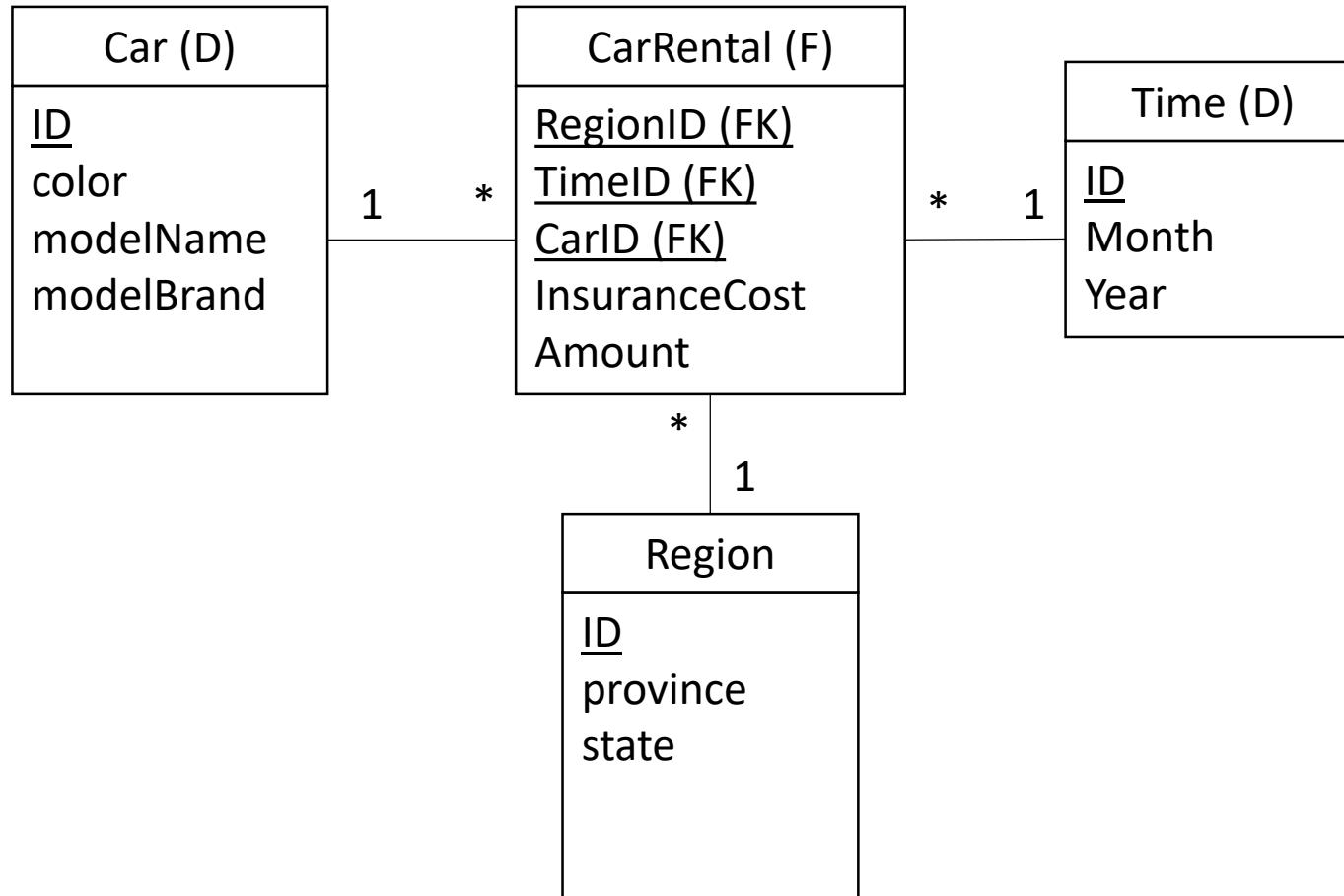
Projection operations are used take the desired fact attributes from the select clause

```

SELECT R.province,
       SUM(insuranceCost),
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
      CR.timeID = T.id AND
      CR.carID = Car.id AND
      R.province = 'Girona'

GROUP BY R.province
ORDER BY T.ID, R.province
  
```

Data cube UML design and transformations



As a final step, we can remove those tables from the FROM clause that are no longer used (i.e., the dimensions we have rolled-up to the all-level)

```

SELECT R.province,
       SUM(amount)
FROM CarRental CR,
     Car C,
     Time T,
     Region R
WHERE CR.regionID = R.id AND
     CR.timeID = T.id AND
     CR.carID = Car.id AND
     R.province = 'Girona'
GROUP BY R.province
ORDER BY R.province
  
```